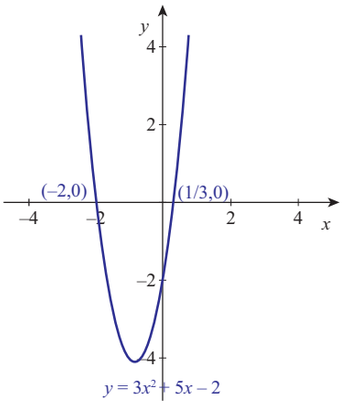


X192B

Figure 2.5

Interval	Sign of $(x+2)$	Sign of $(x-1/3)$	Sign of $3x^2+5x-2$
$(-\infty,-2)$	-	-	+
$(-2,1/3)$	+	-	-
$(1/3,\infty)$	+	+	+

You can see the inequality is satisfied in $[-2,1/3]$.

Example 2.14 Solve $\sqrt{x+14} < x+2$.

Solution:
The function $\sqrt{x+14}$ is defined for $x+14 \geq 0$. Therefore $x \geq -14$, $x+2>0$ implies $x \geq -2$.
$(x+14) < (x+2)^2$ gives $x^2+3x-10>0$.
Hence, $(x+5)(x-2)>0$. Dividing the number line with the critical points $x=-5$ and $x=2$.
Substituting a reference point in the sub-interval we get the solution set to be $x<-5$ and $x>2$.
Since $x \geq -2$, we have the solution to be $x>2$.

Example 2.15 Solve the equation $\sqrt{6-4x-x^2} = x+4$.

Solution:
The given equation is equivalent to the system
$(x+4) \geq 0$ and $6-4x-x^2=(x+4)^2$.
This implies $x \geq -4$ and $x^2+6x+5=0$. Thus, $x=-1,-5$.
But only $x=-1$ satisfies both the conditions. Hence, $x=-1$.

Exercise - 2.5

1. Solve $2x^2+x-15 \leq 0$.
2. Solve $-x^2+3x-2 \geq 0$.